

Paper Directions 27 — my read (Aug 27, 2026)

Source: `transcripts/20260827_Paper_Directions_27_otter_ai_transcript.docx` (~3:46)
Figure shown: `HERO_ew16_allt_min20_mg10_09_21_last_ps_perm_loo_ever_lastps_ew16.png`
Context: Short porch check-in — Charles showed Alex the **09–21 · last-ps · EW16 HERO** after equal-width bar styling (hue by n , counts on bar face) was ported from APGMS/ARGMS minutes plots.
Prior: `PD25_notes.md` · `PD23_notes.md` · `../3-Master_Plan/re_entry/_DISPOSABLE_Alex_hero_population_thread.md` · `../3-Master_Plan/re_entry/SCOUT_and_COMPASS/20260827_SCOUT_to_COMPASS_2009_21_aperture.md`
Process lock (Charles, Aug 27): Transcript captured below. **No code until Charles + COMPASS triage** — some items are **not appropriate yet** (see §Feedback items).

Headline

Alex **loved the figure design** — bar styling, readability, sideways n on the bar face (“very strong”). Two engineering asks: **real x-axis values** in natural equal-width space (not bin index 1–16), and a **clean deck version** without the provenance footer. Two science threads: **positive β_2** (flat, not concave on 09–21) needs interpretation; **lower-echelon bins** deserve diagnostic work (Charles had been focused on the elite tail).

Alex one-liner: “I love the figure overall ... This is a very strong [figure].”

Figure spec (what Alex saw)

Knob	Value
Plot	HERO — mean ever-draft rate vs poolq_loo (LOO teammate perf z)
Binning	EW16 (equal width on observed poolq_loo range)
Seasons	2009–2021 (<code>09_21</code> slug)
Panel	last-ps (one row per athlete, final college season)
Population	ALLT (full panel; not +DFT)
Filters	min20 · mg10 · winsor 0.01–0.99 · PPM z within season
n / K	$n \approx 22,795$ athletes · $K \approx 615$ draftees
LPM readout (on figure)	$\beta_2 \approx +0.00172$ — flat / not concave on this panel

Style elements Alex praised:

- Bar **face color** $\propto$ **bin n** (Blues; sparse bins pale, dense dark)
- Bin population on bar face**, rotated sideways (not floating above)
- Overall **readability** — no more Stack Overflow for label placement

Current gaps Alex flagged:

- X-axis still shows **bin index 1–16**, not **real poolq_loo values**
- n labels: **centered horizontally** on each bar; **bottom-justified** on the bar face (Otter said “left justified” — Charles read: **bottom**, not left)
- Full **provenance strip** needed for lab work; Alex also wants a **presentation-clean** export

Code path: `pass_a_empirical_bundle.py` — `_paint_roster_bars()` when `--poolq-binning equal_width`.

Transcript arc (~3:46)

- X-axis / natural space (00:05–01:15)** — Alex: use real variable values on x , not “bin 1 2 3 ...”; label “average T_j ” (Otter: “T e t j”). Charles: for equal width, bars stay equal width only in **natural space**; quantile would need unequal visual widths if axis is honest. Alex agrees; notes n labels need **bottom justification** on the bar face (may have looked vertically off).
- X-axis reference point (01:15–01:31)** — Open question: when showing **real values** on x , anchor ticks/labels at **bin corner (edge_lo / edge_hi)** vs **bin center**? (Charles read, Aug 27: this is about **axis labeling**, not LPM fit.)
- Style + deck version (01:31–01:53)** — Alex loves it; wants version **without metadata clutter** around the plot.
- β_2 positive (01:53–02:18)** — Charles: LPM quadratic second coefficient **positive** $\rightarrow$ not inverted-U on this panel. Alex: “We are going to have to think about what this means.”
- Lower echelons (02:18–03:20)** — Charles: other plots had **spikes in lowest bins** (sometimes highest bar in bin 1). Alex: “Do a little bit of work on that ... understand what’s happening” — not necessarily data error; could be signal.
- Wrap (03:20–03:40)** — Alex reaffirms love of figure; Charles notes iterative n -label placement (“on the face ... turn him sideways”).

Alex reactions — what landed

Topic	Alex (paraphrase)	Charles / COMPASS read
Overall design	"I love the figure overall ... very strong."	Lock EW styling for deck-quality HERO
Hue + n labels	Cuts matplotlib fiddling; sideways on bar face works	Already shipped; minor alignment tweak pending
Real x-axis	Real values, not bin index; natural EW space	A for EW plots once axis variable confirmed
Axis label "average T_j"*	Said on call	B — figure is poolq_loo HERO, not T_j F-HERO; likely Otter/generic or Alex naming the <i>family</i> of plots
n label alignment	Centered on bar; bottom-justified on bar face, slightly raised from baseline	A — vertical anchor tweak only (horizontal center is correct)
X-axis tick reference	Corner vs center — "think about which is right"	B — where to place real-value x labels/ticks (edge vs midpoint), not LPM anchor
Clean deck version	Strip footer / provenance for slides	A — export flag or sibling PNG
$\beta_2 > 0$ on 09–21	"Think about what this means"	C — science / aperture; no knee chase on this panel yet
Lower-echelon bins	"Do a little bit of work" — spikes, data vs signal	B — diagnostic memo + maybe 1–2 plots; scope with Charles
Data quality	"Data's great ... even better when it's clean"	Supports lower-bin forensics without panic

Alex — key quotes

1. **X-axis**: "Make it actually the **real values** ... don't do bin 1 2 3 4 5 ... make it an x-axis with the **real values**."
2. **EW logic**: "Equal width should be ... **in the natural space**. Otherwise you got to line all these [bars] ... I don't know how they got out of line."
3. **n labels (Charles read)**: Otter: "left justified ... **bottom justified on the bar** ... raise it up just a little bit." → **centered horizontally** on each bar; anchor text at **bottom of bar face** (not floating above). "Left" likely Otter garble.
4. **X-axis reference (Charles read)**: "Do you use the **corner or the center of the bin** as the point of reference ... think about that." → **axis labels/ticks** when switching to real values, not LPM.
5. **Style**: "I love it ... make a version **without all the stuff around it**."
6. **Shape**: "Second coefficient of the squared term is **positive** ... we are going to have to **think about what this means**."
7. **Lower tail**: "You focused a lot on [elite]. **Do a little bit of work on that** [lower echelons] ... let's understand what's happening."
8. **Closing**: "I love the figure overall ... This is a **very strong** [figure]."

Feedback items — triage (no code until Charles approves)

Legend: A = do now (small, locked) · B = discuss first · C = not yet / parked

#	Alex ask	Cat	Notes / blocker
1	Real x-axis values on EW plots (natural space; equal bar widths)	A	Shipped 2026-08-27: bin midpoint tick labels; bars stay categorical width.
2	n label placement — centered on bar, bottom-justified on bar face, slightly raised	A	Shipped 2026-08-27: uniform anchor at bar base (y_pad_frac), not 10% of bar height.
3	Clean deck export — no provenance footer / subtitle clutter	A	e.g. --deck or --no-provenance ; keep full PNG for lab.
4	X-axis tick/label reference — corner (edge) vs bin center	B → locked	Center ticks on figure; edge_lo/edge_hi in roster CSV for forensics. Option B (continuous axis, edges primary) deferred.
5	Axis label — Alex said "average T_j"	B	This figure is poolq_loo ; use correct label per plot family.
6	Lower-echelon diagnostic — spikes in bins 1–3	B	Charles mentioned other sweeps where bin 1 was tallest. Scope: one memo + bin CSV review vs full rerun grid.
7	Interpret $\beta_2 > 0$ on 09–21 last-ps HERO	C	Differs from 13–21 HAND concave; era/aperture science — not a styling ticket.
8	Change population / drop 09–11 / chase concavity on this panel	C	Charles flagged "not appropriate yet" — discuss after lower-bin work.

Suggested first code batch (if Charles green-lights A-items only): #1 + #2 + #3 on EW HERO ew16/ew20 sandbox outputs. Resolve #4 (edge vs center ticks) as part of #1 — **no** LPM or aperture changes.

Naming / object clarity (avoid Alex–Charles drift)

Alex said (Otter)	This figure actually is
"Average T_j "	poolq_loo (LOO teammate mean perf z) — HERO object
"Equal width"	EW16 on poolq_loo range, 09–21 last-ps
Flat / $\beta_2>0$	Marginal roster-context hero — not Alex PD25 **fix-$\hat{A} \times T_j$** knee

F-HERO / CCT (T_j on x) was **not** the figure on screen this call — but Alex’s x-axis ask applies to the whole EW plot family.

How this maps to current campaign

Thread	Link
Working aperture 09–21 last-ps	SCOUT memo 2026-08-27 · K=615
HERO on 09–21	Flat elite ($\beta_2>0$) — not 13–21 HAND concave
F-HERO / P2b overlay	Downturn on **T_j** still visible on 09–21 ALLT (separate figure)
EW vs QTL perm deck	q16/q20/ew16/ew20 on 09–21
APGMS EW styling precedent	Aug 21 BDP — hue + n on bar face

Open / homework

- ☒ Transcript → this doc
- ☐ **Charles:** confirm A/B/C table (especially #4–#8)
- ☐ **COMPASS:** after green-light — implement A-items only; then lower-bin memo (#6)
- ☐ Update disposable thread **YOU ARE HERE** after triage
- ☐ HAND: clean EW HERO for deck vs full provenance for SI

Artifacts

Item	Path
Transcript	transcripts/20260827_Paper_Directions_27_otter_ai_transcript.docx
Figure shown	sports_sandbox/hero/HERO_ew16_allt_min20_mg10_09_21_last_ps_perm_loo_ever_lastps_ew16.png
EW20 sibling	sports_sandbox/hero/HERO_ew20_allt_min20_mg10_09_21_last_ps_perm_loo_ever_lastps_ew20.png
Permutation deck	sports_sandbox/hero_permutation_slides/HERO_permutation_slides_AUTO.pptx
Living population thread	3-Master_Plan/re_entry/_DISPOSABLE_Alex_hero_population_thread.md

Thread log

Date	Entry
2026-08-27 AM	PD27 scaffold; process lock (no code until triage).
2026-08-27 PM	Otter transcript ingested (~3:46). Alex loved EW HERO style. Engineering: real x-axis, n alignment, deck export. Science: $\beta_2>0$ + lower-bin diagnostics — discuss before code .